Transferring Learned Repairs Between Fine-Tunes

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A repair is an extra weight update that aims to reduce a fine-tune's collateral damage while preserving its intended effect. I compare two ways to reuse a repair learned for an earlier fine-tune (the donor): rescale its fixed directions or train from it as an initialization. In a 67M-parameter model, German-degrading fine-tunes also damage Dutch; repairs aim to restore Dutch while retaining the German change. On two independent target fine-tunes at 2,048 fit tokens, donor initialization recovers 98.2% and 98.8% of Dutch damage within a 600-step cap. Rescaling fixed directions recovers 86.9% and 78.4%, even with two additional optimization runs per target that met the declared stopping criteria. Both paired 95% intervals for this advantage lie above the prespecified 3 percentage point (pp) margin. Donor initialization's advantage over fresh fitting is small, meets the prespecified ±2 pp equivalence criterion and is comparable to fresh-seed differences. It is larger in the tested short runs with little data. Among settings meeting the study's preservation constraints and prespecified recovery tolerance, the cheapest donor-initialized setting on the tested grid uses one eighth of the cheapest fresh setting's token presentations. The two families have the same minimum unique fit-token budget. In this setting, the donor repair is more useful as a starting point than as fixed directions.

1 Introduction

A *repair* is an additional weight update trained to reduce fine-tuning's collateral damage while preserving its intended effect. When a recipe is repeated on new samples, a repair learned for an earlier fine-tune, the *donor*, may help repair the next one. Reusing its fixed directions trains only 24 coefficients per target; using it as an initialization allows the repair directions to change.

I test these alternatives by deliberately worsening German next-token prediction, which also raises Dutch cross-entropy (CE), a measure of prediction loss. Dutch is excluded from the fine-tuning objective. In this controlled test, repair must restore Dutch prediction while preserving the German change. The donor and two target fine-tunes are trained independently from one base model, with the same recipe but different seeds and text sampled from different session dates in the same Europarl distribution (Koehn 2005). I use Goodfire's model from its adVersarial Parameter Decomposition (VPD) study (Bushnaq et al. 2026, sec. 3.1). Repairs act on its weights without using the decomposition. Scalar refitting has the same fixed-direction form as coefficient edits of a decomposition (Section 4).

Task arithmetic adds or subtracts a weight difference with a single scale (Ilharco et al. 2023). RESTA reuses a safety vector with a fixed global scale in its main experiments (Bhardwaj et al. 2024), while SafeGene learns a scalar per selected layer for frozen distilled safety adapters (Wang et al. 2026, sec. 3.5 and 4.4). SPoT instead uses learned source-task soft prompts to initialize target-task training (Vu et al. 2022). Here I compare fixed-direction reuse and donor initialization at matched rank on the same targets. Fresh and random starts test the value of donor initialization, and Section 3.3 compares fitting costs.

2 Setup

Model and fine-tunes. The base is a four-layer, 67M-parameter decoder-only model pretrained on a subset of the Pile (Gao et al. 2020). Each fine-tune adds D to the 24 attention and MLP matrices. A hinge term raises German CE while KL keeps English, French, Spanish and Italian, the *retain* languages, close to the base model. Both targets raise Dutch CE by about 0.19 nat/token on held-out evaluation text (method details).

Repair objective and recovery. Each repair adds a rank-one update $E_m = b_m a_m^\top$ to each of the 24 matrices, with input factor a_m and output factor b_m. All repairs obey the same *norm cap*, $\|E\|_F \le \|D\|_F$, where the norms

cover all 24 matrices. The loss sums three equally weighted mean KL divergences $D_{\mathrm{KL}}(p_{\text{teacher}} \parallel p_{\text{repair}})$: to the base model on Dutch positions, rewarding restoration of Dutch predictions, and to the fine-tuned model on German and retain positions, penalizing reversal of the German effect or changes to retain predictions. Recovery measures the share of Dutch CE damage removed by repair X:

$$r_{\mathrm{nl}}(X) = 1 - \frac{\mathrm{CE}_{\mathrm{nl}}(X) - \mathrm{CE}_{\mathrm{nl}}(\text{base})}{\mathrm{CE}_{\mathrm{nl}}(\text{fine-tuned}) - \mathrm{CE}_{\mathrm{nl}}(\text{base})}.$$

Recovery differences are in percentage points (pp).

Families. One donor repair, learned from fresh initialization on 73,728 prediction positions, serves both targets. Scalar-only fitting learns coefficients in $E_m = c_m b_m a_m^\top$ over its fixed factors. Other families train one or both factors (Table 1). Applying the capped donor repair without target fitting gives the *fixed donor repair* reference.

Table 1: Families share rank one per matrix, but train different numbers of parameters. Donor starts are scaled down if needed to meet the cap. Fresh follows LoRA's initialization (Hu et al. 2022).

Family	Start	Trained (parameters)	What the comparison tests
Scalar coefficients only	capped donor repair	coefficients c_m (24)	recovery by rescaling fixed directions
Donor initialization	capped donor repair	both factors (55,296)	gains from training directions and using the donor start
Donor initialization, output only	capped donor repair	output vectors b_m (27,648)	recovery with donor input vectors fixed
Random initialization	random directions, donor-start norm	both factors (55,296)	benefit of donor directions beyond starting norm
Fresh	$b_m = 0$, $a_m \sim \mathcal{N}(0, 1/d_{\mathrm{in}})$	both factors (55,296)	benefit of donor training over fitting anew

The first fresh seed uses the random input vectors from which the donor repair was trained. Its comparison with donor initialization assesses donor training as a whole, without isolating directions, norm or checkpoint selection. The second fresh seed uses an independent draw. Second seeds of fresh and random initialization were tested only at 2,048 fit tokens with a 600-step cap.

Budgets and step caps. A fit budget counts the unique prediction positions (*fit tokens*) used for fitting: 256, 512 or 2,048, nested so that larger budgets add text to smaller ones. They exclude a shared selection and diagnostic pool of 57,344 positions. All families and both targets share fitting, selection and evaluation windows. Ordinary fitting runs full-batch Adam at four learning rates per family, with a 10-step warm-up and then a constant rate, saving a checkpoint every 50 steps. A step cap of 50, 100, 200 or 600 limits selection to checkpoints saved up to that step, so the chosen checkpoint can precede the cap (fitting details). The *full budget* is 2,048 fit tokens with a 600-step cap for ordinary runs.

Selection and readout. Evaluation text is grouped by Europarl session date, with one 256-prediction window per language per date. For each family, budget and cap, selection on 32 dates separate from fitting favors Dutch recovery among candidates preserving the German effect and reducing Dutch KL from the base, with a guard on retain-language CE (full rule). Selected repairs are evaluated once on 92 other validation-period dates, the *readout*, separate from earlier project windows. The *readout constraints* recheck German-effect preservation, positive Dutch KL recovery and retain-language CE drift there (thresholds in Table 6). Within each fixed target, one joint date bootstrap (Efron 1979) gives paired 95% percentile intervals.

Primary comparison. Donor initialization is compared with scalar-only fitting at the full budget. To check whether the 600-step cap stops scalar fitting too early, the *expanded scalar search* adds two projected-gradient runs on the same loss from two starts, so optimization budgets are unequal. Selection pools their final candidates with the 48 ordinary checkpoints (details).

An advantage above 3 pp is established when the whole paired interval exceeds 3 pp, excluded when wholly below, and undecided otherwise. Interpretation requires measurable Dutch damage, both selected repairs

meeting the readout constraints, and both additional scalar runs meeting the declared stopping criteria (Table 6; checks). Other contrasts are descriptive, without multiplicity correction. Their intervals determine whether the sign is positive, negative or undecided. Five contrasts in Table 3 also assess equivalence within ±2 pp. The protocol fixed both margins without a stated derivation.

3 Results

3.1 Fixed directions fall short

Without target fitting, the capped donor repair recovers 86.3% and 58.6% of Dutch damage on the primary and replication targets. At 2,048 fit tokens, the expanded scalar search raises recovery to 86.9% and 78.4%, while training both factors from the donor start reaches 98.2% and 98.8% (Figure 1).

Both additional scalar runs on each target met the declared stopping criteria (Table 7). They left selection unchanged on the primary target and raised replication recovery from 77.3% to 78.4% (Table 6).

Both targets meet the conditions for interpreting the primary contrast (Table 6 and Table 7). Donor initialization exceeds the expanded scalar search by 11.3 pp [+10.0, +12.7] on the primary target (Table 2) and 20.4 pp [+19.0, +21.7] on the replication target. Both intervals lie entirely above the prespecified 3 pp margin.

The scalar shortfall is already present on the fitting data, so it is not confined to new text. At the full budget, the selected scalar repairs remove 87.0% and 79.2% of the fine-tune's Dutch KL from the base model, against at least 98.4% for every factor-training family. This measures KL recovery, whereas readout uses CE. The bounded search cannot distinguish limitations of the fixed directions under the norm cap from remaining optimization error.

At the full budget, training only the output vectors from the donor start closes most of the gap: output-only fitting recovers 10.2 and 19.1 pp more than the expanded scalar search on the primary and replication targets, and training both factors adds 1.1 and 1.3 pp (Table 3). Without a frozen-random-input control, the benefit of the donor's learned input vectors remains unresolved.

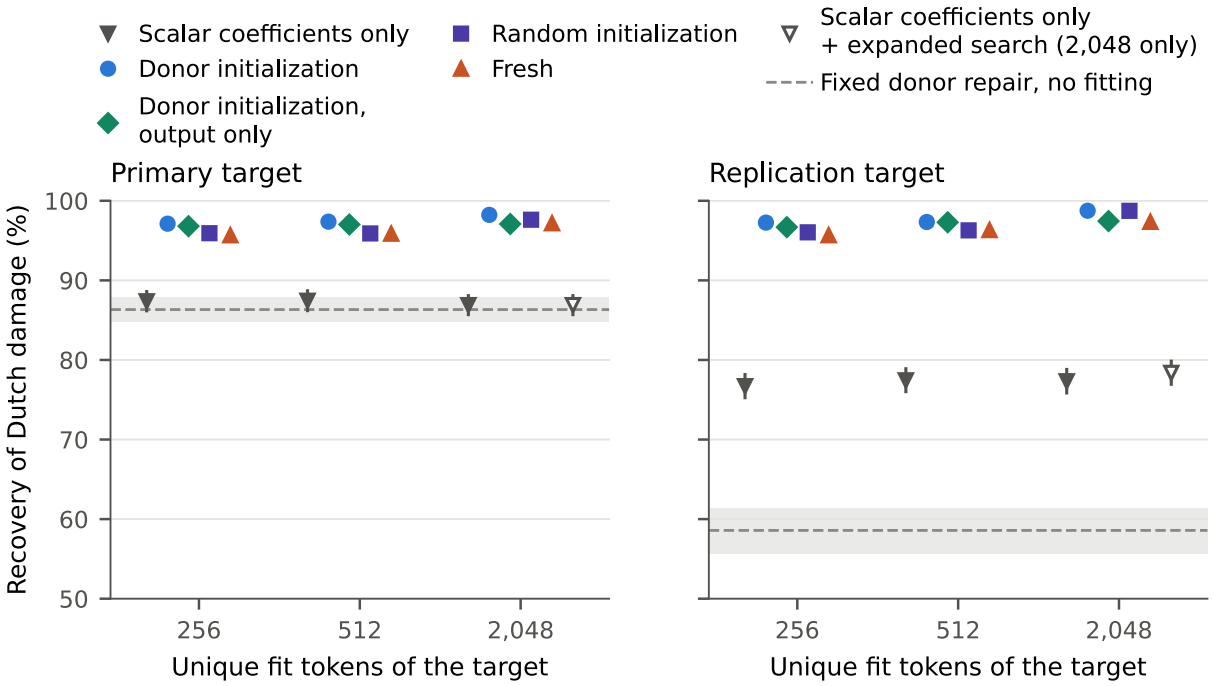


Figure 1: Dutch-damage recovery with a 600-step cap for ordinary runs. Factor-training families recover at least 95.6% across all three fit budgets. Whiskers and the capped fixed-donor band show 95% bootstrap intervals over 92 dates. At 2,048 tokens, hollow markers show selection after adding two additional candidates that met the declared stopping criteria. No additional scalar runs were made at 256 or 512 tokens.

Table 2: Primary target, 2,048 fit tokens, 92 readout dates. 95% intervals from a joint bootstrap over dates (5,000 draws). Donor-initialized fitting exceeds the tested scalar-coefficient search by more than 3 percentage points. The scalar search includes 48 checkpoints from 4 learning-rate runs with a 600-step cap and two final candidates from additional full-batch runs that met the declared stopping criteria. Selection considers all 50 candidates and can retain an ordinary checkpoint.

Repair	Recovery of Dutch damage (%)	95% CI
Donor initialization (both factors trained)	98.2	[97.6, 98.9]
Scalar coefficients only + expanded search	86.9	[85.5, 88.3]
Difference (percentage points)	+11.3	[+10.0, +12.7]

3.2 The donor as an initialization

Fresh and norm-matched random starts test how much the donor contributes as an initialization. At the full budget, its advantages over the first fresh seed are +1.1 and +1.4 pp on the primary and replication targets. Both intervals are positive and inside the prespecified ±2 pp equivalence band (Table 3). Differences between fresh seeds are similar in size. Against the independent second fresh seed, the advantages shrink to +0.04 and +0.7 pp, and the primary-target interval includes zero.

Against the norm-matched random start, the advantage is +0.6 pp on the primary target and +0.01 pp [−0.26, +0.28] on the replication target. The latter's sign is undecided. Equivalence was not assessed for this contrast.

With 256 fit tokens and a 50-step cap, donor initialization leads fresh fitting by +4.4 and +4.1 pp and the random start by +4.9 and +4.5 pp (Figure 2). This larger advantage is limited to the tested starts and fitting windows: one seed per control, with shared fitting windows across targets.

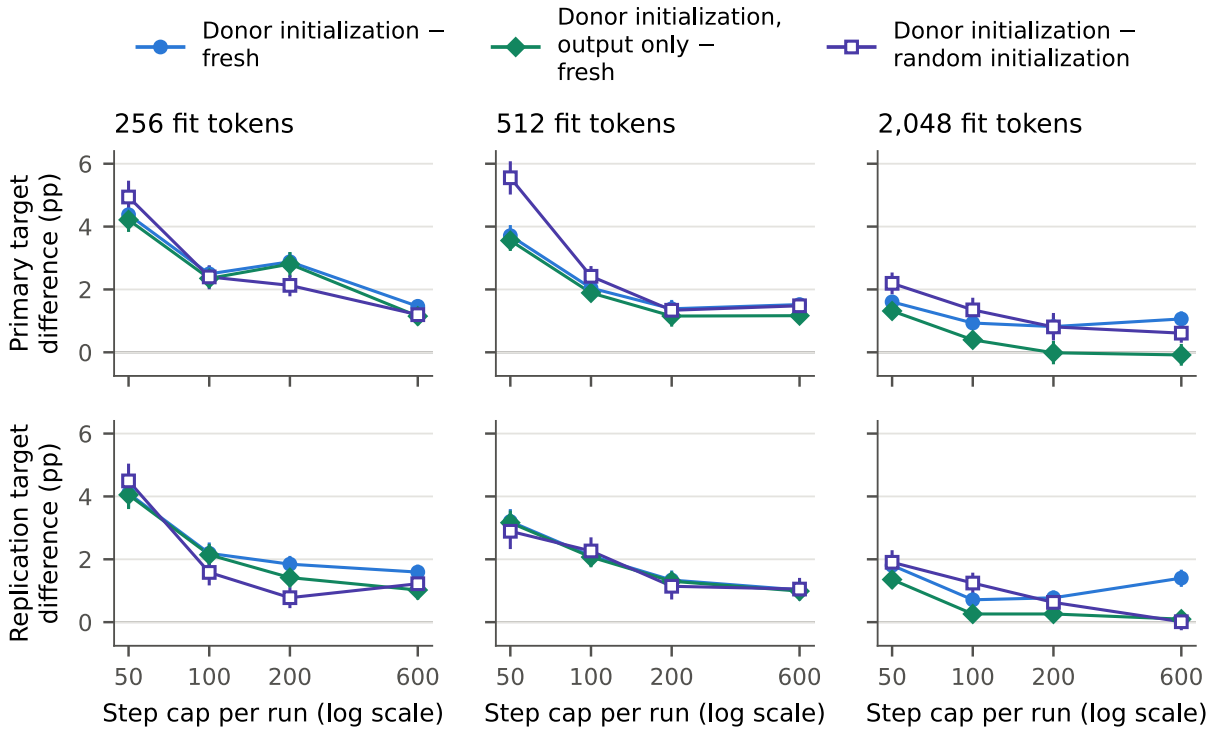


Figure 2: Paired recovery differences at matched fit budgets (columns) and step caps (logarithmic axis), with 95% bootstrap intervals over 92 dates per target. Primary and replication targets occupy the top and bottom rows. The tested donor start generally helps most with little data and few steps.

Table 3: Secondary contrasts at the full fit budget (2,048 unique fit tokens, 600-step cap for ordinary runs): paired difference in recovery (pp) with its 95% interval, 92 readout dates per target. For rows marked '+ expanded search', scalar-repair selection also considers two final candidates from additional runs that met the declared stopping criteria. These intervals are descriptive, without multiplicity correction. The last column applies only to the five contrasts named in the protocol for an equivalence reading (primary / replication): 'yes' means that the whole interval lies inside the ±2 pp band.

Contrast (2,048 fit tokens)	Primary target	Replication target	Inside ±2 pp
Donor initialization, output only − scalar coefficients only + expanded search	+10.2 [+8.9, +11.5]	+19.1 [+17.6, +20.5]	—
Donor initialization − donor initialization, output only	+1.1 [+0.7, +1.5]	+1.3 [+1.0, +1.6]	yes / yes
Donor initialization − fresh	+1.1 [+0.8, +1.3]	+1.4 [+1.1, +1.7]	yes / yes
Donor initialization, output only − fresh	−0.1 [−0.4, +0.3]	+0.1 [−0.2, +0.4]	—
Donor initialization − random initialization	+0.6 [+0.3, +0.9]	+0.01 [−0.26, +0.28]	—
Random initialization − fresh	+0.5 [+0.2, +0.7]	+1.4 [+1.0, +1.7]	yes / yes
Scalar coefficients only + expanded search − fresh	−10.3 [−11.7, −8.9]	−19.0 [−20.4, −17.5]	—
Scalar coefficients only + expanded search − fixed donor repair	+0.6 [+0.1, +1.1]	+19.8 [+17.8, +22.0]	—
Scalar coefficients only + expanded search − scalar coefficients only	same repair selected	+1.1 [+0.8, +1.3]	—
Fresh, second seed − fresh	+1.0 [+0.7, +1.3]	+0.7 [+0.4, +0.9]	yes / yes
Random initialization, second seed − random initialization	−0.38 [−0.76, −0.02]	−1.2 [−1.5, −0.9]	yes / yes
Donor initialization − fresh, second seed	+0.04 [−0.19, +0.26]	+0.7 [+0.5, +1.0]	—

3.3 What donor initialization saves

Each combination of family, fit budget and step cap is a *cell*. The reference is the first fresh seed at the full budget. A cell *reaches* it if the selected repair satisfies the readout constraints and the lower bound of its paired 95% interval for recovery minus reference recovery is strictly above −2 pp. Presentations count fit tokens × optimizer steps over all four learning-rate runs. The cheapest reaching donor and fresh cells use 256 and 2,048 tokens, respectively, with 50-step caps: 51,200 versus 409,600 presentations, one eighth as many. Fresh fitting also reaches with 256 tokens at a 600-step cap, so both families have the same minimum of 256 unique fit tokens, or 57,600 with the shared 57,344 selection positions. This saves presentations without reducing the unique-token requirement.

Training the donor repair took 3,686,400 presentations, so this per-target difference would repay its cost from the eleventh target, a projection from two tested targets. The estimated GPU-time saving is much smaller (Table 4), moving break-even to about 151 and 106 targets for the primary and replication targets, without establishing an end-to-end time saving. Checkpoints are selected before readout, but the cheapest reaching cells were chosen afterward, and the intervals describe cells, not the cost ratio. These estimates depend on the reference seed, the one-sided reach threshold and the tested grid (cost accounting).

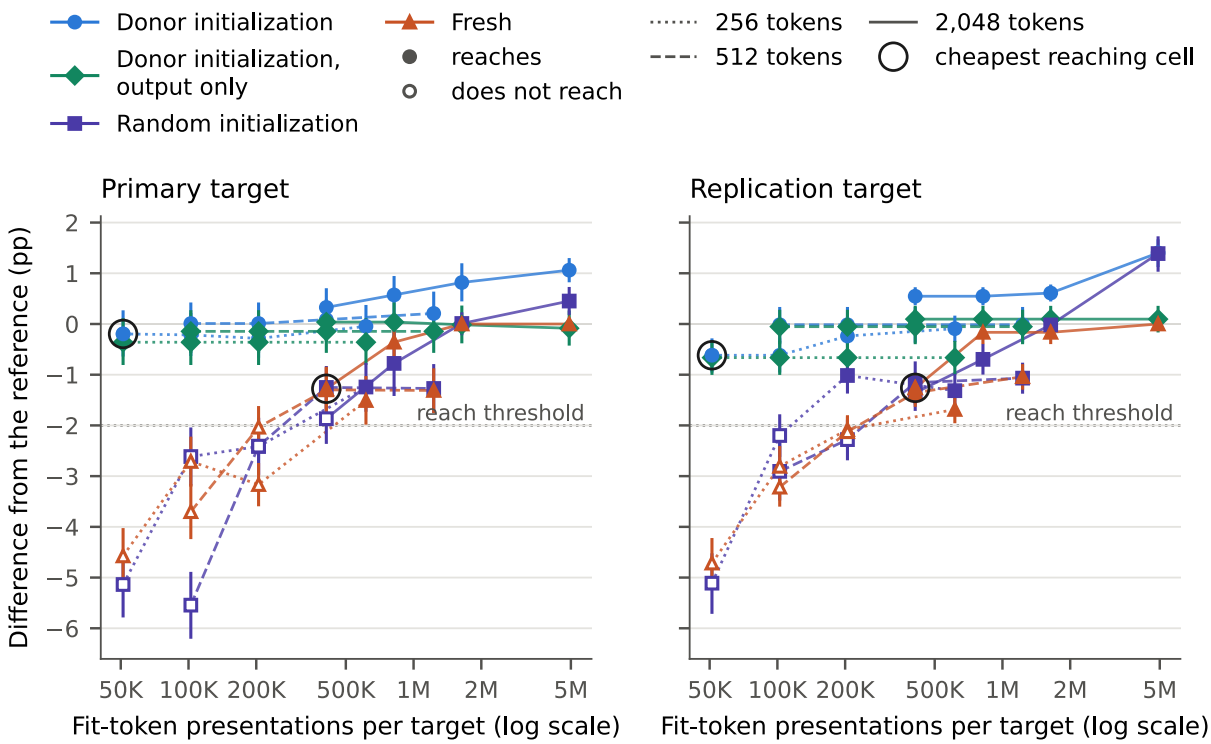


Figure 3: Recovery difference from the first fresh seed at 2,048 tokens and a 600-step cap versus fit-token presentations, with paired 95% bootstrap intervals over 92 dates per fixed target. Lines join step caps 50, 100, 200 and 600. Filled markers satisfy readout constraints and have interval lower bounds strictly above −2 pp. Rings mark the cheapest reaching settings: donor initialization at 256 tokens and fresh at 2,048, both with a 50-step cap. Scalar-only cells never reach and are omitted.

Table 4: Target-side cost of the cheapest cell of fresh fitting and of donor initialization that reaches the reference (fresh at 2,048 tokens and up to 600 steps), found separately in each currency. A cell is written as unique fit tokens / step cap. K* = donor cost / (fresh cost − donor-initialization cost) is a modelled amortization: reuse pays off for more than K* targets. The donor's one-time cost and what each currency counts and leaves out are in the cost-accounting appendix. GPU seconds are an estimate of allocated shared work, and the donor's come from a different run.

Currency	Target	Fresh	Fresh cell	Donor init.	Donor init. cell	Break-even K*
Unique tokens	Primary	57,600	256 / 600	57,600	256 / 600	no per-target saving
	Replication	57,600	256 / 600	57,600	256 / 600	no per-target saving
Fit-token presentations	Primary	409,600	2,048 / ≤50	51,200	256 / ≤50	10.3
	Replication	409,600	2,048 / ≤50	51,200	256 / ≤50	10.3
GPU seconds (estimate)	Primary	16.8	2,048 / ≤50	14.5	256 / ≤50	151.4
	Replication	17.1	2,048 / ≤50	13.9	256 / ≤50	105.9

4 Discussion

Rescaling the donor's fixed directions recovered 86.9% and 78.4% of Dutch damage after target fitting. Training the rank-one factors recovered at least 95.6% across the tested starts and fit budgets with a 600-step cap. The donor start added little over fresh fitting at the full budget, but helped more in the tested short runs: its cheapest reaching cell used one eighth of fresh fitting's token presentations. The two families had the same minimum

unique fit-token budget. In this setting, the donor repair was more useful as a starting point than as a set of fixed directions.

The scalar shortfall is already visible on the scalar fits' own fitting data. Coefficient edits of a parameter decomposition such as VPD (Bushnaq et al. 2026) also rescale fixed rank-one components. As an interpretation not tested here, fitting such coefficients on the edit's own data could flag a shortfall before evaluation on new text, although it would not distinguish limitations of the components from optimization error under a given search and norm cap.

5 Limits of these conclusions

These results concern one donor, one recipe and one model, with two targets analysed separately. Date-bootstrap intervals cover variation across readout dates for fixed repairs, not across fine-tunes, donors, recipes or fitting windows. Second seeds were tested only at the full budget, leaving short-run results conditional on the tested starts and shared fitting windows. I do not analyse what the repair directions represent or how they restore Dutch prediction. Without an output-only control with frozen random input vectors, the donor input vectors' contribution is also unresolved; Zhu et al. (2024) studied such frozen random inputs. The study does not test new-skill transfer or safety alignment, including the safety settings of RESTA and SafeGene.

AI assistance. I used OpenAI Codex and Anthropic Claude for research planning, code development, data analysis and manuscript preparation. Scientific Agent Skills (Kassis et al. 2026) supplied the scientific-writing, scientific-critical-thinking, peer-review and citation-management workflows used to review organization, claims and references.

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Appendix A — Method details

Fine-tunes. Adam (Kingma and Ba 2015) learns full-matrix additive updates D_m to each layer's query, key, value, output and two MLP matrices (24 total), with embeddings and norms frozen. It runs at learning rate 3e-6 for 200 steps on the per-window hinge $\max(0, 1 - \Delta\mathrm{CE}_{\mathrm{de}})$ (1-nat margin) plus retain-language KL with weight 1. Checkpoints are selected on German and retain-language validation metrics.

Repairs. Ordinary fitting uses Adam. The donor starts fresh (Table 1), using 73,728 prediction positions, three learning rates and 200 minibatch steps per run. Checkpoints saved every 25 steps undergo selection on donor validation windows by the rule below. Targets save checkpoints every 50 steps at four learning rates (Table 5): 48 candidates per family at the 600-step cap. Donor starts are capped as $\min(1, \|D\|_F / \|E_{\mathrm{donor}}\|_F)\, E_{\mathrm{donor}}$.

Table 5: Learning-rate grids of the repair families.

Family	Learning rates
Scalar coefficients only	3e-3, 1e-2, 3e-2, 1e-1
Donor initialization, donor initialization output only, random initialization and fresh	1e-4, 3e-4, 1e-3, 3e-3

The 50-step cap uses the same grid. Several output-only cells and donor-initialized replication at 512 tokens select the first checkpoint at the lowest rate, which can explain repeated values across caps. Advantages over fresh and random starts are nonmonotonic in budget and cap.

Selection. Feasibility requires a German-damage ratio (with repair / without repair) of 0.9–1.1, German KL drift from the fine-tune at most 0.05 times the base model's KL from it, and Dutch movement toward the base. Feasible candidates rank by Dutch recovery, then smaller norm, then listing order. Of the top five, selection takes the first raising no retain language's CE by more than 0.01 nat/token.

Data. Fit budgets exclude each window's initial context token and split predictions 2:1:1 across Dutch, German and retain text. At 256 tokens, one Dutch window contributes 128 predictions, one German 64, and four retain windows 16 each. At 512 tokens these counts double. At 2,048, four Dutch and two German windows contribute 256 each, and one per retain language contributes 128. Target fit windows and selection dates are separate from the donor's.

Selection and readout use 32 and 92 dates, respectively, with one 257-token window (256 predictions) per language from the same turn. Portuguese, the seventh language, is diagnostic only: its selection windows count toward token and evaluation costs, but not the loss or selection constraints. Readout dates were ineligible if present in recorded project data, pilot or run files, or if their windows duplicated text in use. All passed. Overlap with the base model's pretraining text remains unknown.

Assessability. The primary contrast requires a Dutch-damage readout interval with lower bound at least 0.025 nat/token. Damage is 0.187 nat/token [0.178, 0.196] on the primary target and 0.189 [0.180, 0.198] on replication. Both selected repairs must satisfy the readout constraints (Table 6), and both additional scalar runs must meet the stopping criteria.

Study record. My records place a dated protocol fixing rules, margins and families before readout-data construction and target fine-tuning. An author-dated amendment fixed readout data and clarified the primary interpretation before any target fine-tune or readout, without changing decision rules. The unpublished protocol, amendment and log lack independent timestamps. This is not verifiable preregistration.

Uncertainty. Within each target, 5,000 joint date resamples cover all budgets and caps, computing recovery ratios from mean damages. Readout constraints use point estimates over all dates. Equivalence is assessed only

for the five contrasts marked in Table 3. It remains unestablished between donor and random initialization at every budget and cap.

Appendix B — Checks behind the primary reading

All 2,048-token cells and the fixed donor repair satisfy the readout constraints (Table 6).

Table 6: Readout of the 2,048-token cells (600-step cap for ordinary runs) and the fixed donor repair on the 92 readout dates. Rows marked '+ expanded search' report the repair selected from ordinary checkpoints and two final candidates from additional runs that met the declared stopping criteria. KL recovery (%) is the share of the fine-tune's Dutch KL from the base model that the repair removes (required to be positive). German effect kept is the ratio of German CE damage with and without the repair (required 0.9–1.1). German drift is the repair's German KL from the fine-tuned model relative to the base model's KL from the fine-tuned model (required at most 0.05). Worst retain change is the largest CE change in English, French, Spanish or Italian against the fine-tuned model in nat/token (required at most +0.01).

Target	Repair	Recovery (%) [95% CI]	KL recovery	German effect kept	German drift	Worst retain change
Primary	Scalar coefficients only	86.9 [85.5, 88.3]	86.7	1.005	0.0003	+0.0003
Primary	Scalar coefficients only + expanded search	86.9 [85.5, 88.3]	86.7	1.005	0.0003	+0.0003
Primary	Donor initialization	98.2 [97.6, 98.9]	98.3	0.996	0.0005	−0.0007
Primary	Donor initialization, output only	97.1 [96.4, 97.7]	98.1	1.003	0.0003	−0.0006
Primary	Random initialization	97.6 [97.0, 98.3]	98.2	0.988	0.0005	−0.0009
Primary	Random initialization, second seed	97.2 [96.5, 98.0]	97.5	0.990	0.0005	−0.0012
Primary	Fresh	97.2 [96.5, 97.8]	97.7	0.990	0.0005	−0.0014
Primary	Fresh, second seed	98.2 [97.5, 98.9]	97.8	0.985	0.0006	−0.0015
Primary	Fixed donor repair	86.3 [84.8, 87.9]	84.4	0.999	0.0002	−0.0001
Replication	Scalar coefficients only	77.3 [75.7, 79.0]	76.5	1.007	0.0002	+0.0003
Replication	Scalar coefficients only + expanded search	78.4 [76.7, 80.0]	77.8	1.007	0.0003	+0.0004
Replication	Donor initialization	98.8 [98.1, 99.4]	98.6	1.001	0.0003	−0.0005
Replication	Donor initialization, output only	97.4 [96.8, 98.1]	98.2	1.007	0.0002	−0.0006
Replication	Random initialization	98.7 [98.0, 99.4]	97.9	0.990	0.0004	−0.0010
Replication	Random initialization, second seed	97.5 [96.8, 98.3]	97.8	0.996	0.0004	−0.0009
Replication	Fresh	97.4 [96.7, 98.0]	98.1	0.994	0.0004	−0.0012
Replication	Fresh, second seed	98.0 [97.4, 98.6]	98.4	0.990	0.0004	−0.0012
Replication	Fixed donor repair	58.6 [55.6, 61.4]	48.0	1.009	0.0003	−0.0001

The additional scalar search uses $e_m = c_m \|b_m\| \|a_m\|$, so $\|E\|_F = \|e\|_2$ and radial scaling projects exactly onto the norm cap. It combines the spectral step of Barzilai and Borwein (1988), the projected-gradient scheme of Birgin et al. (2000) and monotone Armijo line search, with at most 300 iterations from each of two starts: initial coefficients and the lowest-fit-loss checkpoint.

The declared stopping criteria require finite values for all losses and gradients, including line-search trials, initial training/export loss agreement within 1e-5 nat/token, and at least one stopping condition: projected-gradient norm at most 1e-4 of its reference; relative fit-loss decrease below 1e-7 for five iterations; or the line-search floor with stationarity at most 1e-3 of that reference. Both starts use the initial coefficients' projected-gradient reference. Final-loss agreement is descriptive, not a stopping criterion.

Table 7: Both starts met the declared stopping criteria on both targets. End losses differ by a relative 2.0e-5 (primary) and 1.1e-6 (replication).

Target	Start	Stop	Iterations	Fit loss, start → end	Stationarity / reference
Primary	initial scalar coefficients	relative decrease	107	0.02516 → 0.02070	0.0062
Primary	best checkpoint	relative decrease	84	0.02129 → 0.02070	0.0019
Replication	initial scalar coefficients	line-search floor	69	0.08323 → 0.03682	0.0002
Replication	best checkpoint	relative decrease	49	0.03870 → 0.03682	0.0004

The search covers four learning rates for 600 steps plus two further starts under $\|E\|_F \le \|D\|_F$. Other assignments of the 24 coefficients and larger repair norms remain untested.

Appendix C — Cost accounting

For each cost measure, let C^* be a family's cheapest reaching cost on the tested grid. If the observed per-target difference repeats, reuse repays the donor cost after more than

$$K^* = \frac{C_{\text{donor}}}{C^*(\text{fresh}) - C^*(\text{donor initialization})}$$

targets (Table 4). The donor's 3,686,400 presentations give $K^* \approx 10.3$: an 11-target projection from two targets. Neither scalar-only fitting nor the fixed donor repair reaches the reference (Figure 3). A random start saves no presentations on the primary target. On replication it gives $K^* = 18$, charged with the full donor cost despite using only the donor repair's norm.

Unique tokens count fit tokens plus 57,344 selection positions (32 dates × 7 languages × 256). Presentations count fit tokens × optimizer steps across all four learning rates up to the cap, excluding selection forward passes. GPU seconds combine training time prorated to the cap and selection-evaluation time allocated by candidate count. For the expanded scalar search, presentations and GPU seconds also include both additional runs and evaluation of the 48 checkpoints used to choose the second start. All cells exclude base/fine-tuned readouts, the pre-fitting damage check and exploration of other cells.

The donor costs 131,072 unique tokens (73,728 fit plus selection), 3,686,400 presentations (3 learning rates × 200 steps × 6,144 tokens per minibatch), and an estimated 336 GPU seconds for training, selection and readout. Its fine-tune is excluded because the donor model is assumed to exist.

The first fresh seed is the reference (Section 3.3), with self-comparison interval [0, 0]. At 256 tokens and a 600-step cap, fresh fitting's lower bounds, −1.985 pp (primary) and −1.957 pp (replication), narrowly exceed −2 pp and set its lowest unique-token cost. Sensitivity to another reference seed remains unevaluated.

Appendix D — Reproducibility and availability

Code, saved measurements, and instructions for replaying analysis and running the model procedures are on GitHub.